Sahaj Bhadja

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Summary

Software Engineer and USC MSCS graduate with 2+ years building **Java and Python data systems**, ETL pipelines, data models, SQL services, automated tests, and cloud-native deployments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Data Structures, Analysis of Algorithms, Database Systems, Software Engineering

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Web Technologies, Computer Networks, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Built normalized Azure SQL schemas, indexed queries, and ETL-style integrations processing 5K+ monthly jobs, reducing dashboard latency from 1.2 seconds to 350 milliseconds for 500+ **data platform users**.
- Reworked asynchronous Python services with FastAPI workers, explicit job states, retries, and failure isolation, tripling throughput for 1K+ concurrent jobs while sustaining 99%+ **pipeline reliability** in production.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Developed concurrent Python pipelines for secure file ingestion and multimodal transformation, processing 1K+ files per batch with provider fallbacks, validation, and **automated integration tests** across production workflows.
- Built Flask services and REST APIs supporting 10K+ daily requests, integrating AWS RDS data models and CI/CD checks while reducing **data processing turnaround** 60% during peak workloads.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built reproducible Python pipelines processing 100K+ autonomous-driving trajectory logs, improving planning 18% through structured **large-dataset transformation**, feature preparation, controlled evaluations, and documented baseline comparisons for scalable research analysis.
- Architected a spatial-temporal encoder combining HD-map geometry, lane context, and ego trajectories, improving closed-loop stability 15% through **data modeling**, debugging, experiment reviews, and systematic validation across simulation experiments.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Engineered object-oriented integration services and ETL workflows using AWS Glue, S3, and REST APIs, improving pipeline efficiency 68%, response times 30%, and **distributed data reliability** across platforms.
- Automated 20+ unit and integration tests with PyTest, Mockito, Jenkins, and virtual-machine builds, supporting **code reviews**, debugging, Git collaboration, and dependable releases across Windows, Linux, and macOS.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built Databricks and Spark microbatch pipelines that cleaned, joined, and transformed high-volume service records, enabling **reproducible data processing** and operational reporting across 50K+ records with sub-200-millisecond retrieval.
- Designed SQL data models and integration contracts for 12+ endpoints, translating business requirements into documented schemas while reducing **data design errors** 25% before implementation and stakeholder review.

Skills

Languages: Java, Python, SQL, XML, JavaScript, TypeScript, C++, Bash, HTML5, CSS3
Data Engineering: Apache Kafka, Confluent Kafka, Apache Airflow, Databricks, Apache Spark, Apache Iceberg, ETL, Data Pipelines, Data Modeling, RDF, JSON-LD, Linked Data, PostgreSQL, SQL Server, MySQL, MongoDB
Backend & Cloud: Spring Boot, FastAPI, Flask, REST APIs, Microservices, Kubernetes, Docker, Terraform, AWS, Azure, Linux, Unix, Distributed Systems, Concurrency
Engineering: Git, GitHub, GitHub Actions, CI/CD, Jenkins, Unit Testing, Integration Testing, PyTest, Mockito, JUnit, Code Review, Debugging, Agile, Waterfall, SDLC, Technical Documentation, Performance Optimization

Projects

ResearchGraph Harvester | *Java, Spring Boot, Kafka, RDF, JSON-LD, Kubernetes*

- Designed a Java ingestion service that harvested heterogeneous research datasets through Kafka, normalized records into **RDF and JSON-LD data models**, validated provenance, and exposed documented Spring Boot APIs.

Dataset Lakehouse Pipeline | *Python, Airflow, Databricks, Iceberg, Terraform, GitHub Actions*

- Built a Python pipeline orchestrated with Airflow to transform partitioned datasets in Databricks and Apache Iceberg, using **Terraform and GitHub Actions** for reproducible infrastructure, testing, and deployment.

Chat2SQL | *Java, Spring Boot, SQL, PostgreSQL, Docker, Maven*

- Extended a cross-platform Java workspace supporting 10+ databases with schema exploration, query generation, Docker tooling, and Maven builds, applying **relational data modeling** across reproducible Windows, Linux, and macOS environments.